Chapter 6: Problems 6.14, 6.17, 6.22, 6.25, 6.29, 6.32, 6.35, 6.40, 6.46

*6.14 In the circuit of Fig. P6.12, $V_0 = 12$ V, $R_1 = 0.4\ \Omega$, $R_2 = 1.2\ \Omega$, and $L = 0.1$ H. What should the value of C be in order for $i_L(t)$ to exhibit a critically damped response? Provide an expression for $i_L(t)$ and plot its waveform for $t \geq 0$.

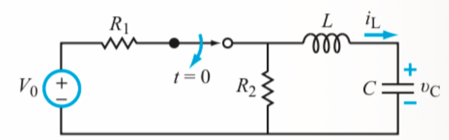
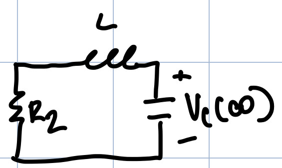


Figure P6.12 Circuit for Problems 6.12 to 6.14.

critical → $a = W_0$

$\frac{R_2}{2L}$ $\frac{1}{\sqrt{LC}}$

$t > 0$



$a = \frac{R}{2L}$

$W_0 = \frac{1}{\sqrt{LC}}$

Solve for C

$$\left(\frac{R_2}{2L}\right)^2 = \left(\frac{1}{\sqrt{LC}}\right)^2$$

$$\frac{R_2^{\,2}}{4L^2} = \frac{1}{LC}$$

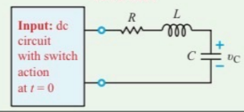
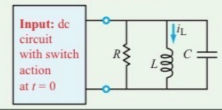
↓

inverse

$$\frac{4L^2}{R_2^{\,2}} = LC \rightarrow C = \cancel{L}\frac{4L^{\cancel{2}}}{R_2^2} \rightarrow \frac{4L}{R_2^{\,2}} = \frac{4(.1[H])}{(1.2)^2[\Omega]}$$

$$= .28[C]$$

Table 6-1: Step response of RLC circuits for $t \geq 0$.

Series RLC	Parallel RLC
	
Total Response	**Total Response**
Overdamped ($\alpha > \omega_0$)	**Overdamped** ($\alpha > \omega_0$)
$v_C(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t} + v_C(\infty)$	$i_L(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t} + i_L(\infty)$
$A_1 = \frac{\frac{1}{C} i_C(0) - s_2[v_C(0) - v_C(\infty)]}{s_1 - s_2}$	$A_1 = \frac{\frac{1}{L} v_L(0) - s_2[i_L(0) - i_L(\infty)]}{s_1 - s_2}$
$A_2 = \frac{\frac{1}{C} i_C(0) - s_1[v_C(0) - v_C(\infty)]}{s_2 - s_1}$	$A_2 = \frac{\frac{1}{L} v_L(0) - s_1[i_L(0) - i_L(\infty)]}{s_2 - s_1}$
Critically Damped ($\alpha = \omega_0$) $i_C = C\frac{di}{dt}$	**Critically Damped** ($\alpha = \omega_0$)
5 $v_C(t) = (B_1 + B_2 t)e^{-\alpha t} + v_C(\infty)$	$i_L(t) = (B_1 + B_2 t)e^{-\alpha t} + i_L(\infty)$
3 $B_1 = v_C(0) - v_C(\infty)$	$B_1 = i_L(0) - i_L(\infty)$
4 $B_2 = \frac{1}{C} i_C(0) + \alpha[v_C(0) - v_C(\infty)]$	$B_2 = \frac{1}{L} v_L(0) + \alpha[i_L(0) - i_L(\infty)]$
Underdamped ($\alpha < \omega_0$)	**Underdamped** ($\alpha < \omega_0$)
$v_C(t) = e^{-\alpha t}(D_1 \cos\omega_d t + D_2 \sin\omega_d t) + v_C(\infty)$	$i_L(t) = e^{-\alpha t}(D_1 \cos\omega_d t + D_2 \sin\omega_d t) + i_L(\infty)$
$D_1 = v_C(0) - v_C(\infty)$	$D_1 = i_L(0) - i_L(\infty)$
$D_2 = \frac{\frac{1}{C} i_C(0) + \alpha[v_C(0) - v_C(\infty)]}{\omega_d}$	$D_2 = \frac{\frac{1}{L} v_L(0) + \alpha[i_L(0) - i_L(\infty)]}{\omega_d}$
Auxiliary Relations	
1 $\alpha = \begin{cases} \frac{R}{2L} & \text{Series RLC} \\ \frac{1}{2RC} & \text{Parallel RLC} \end{cases}$	$\omega_0 = \frac{1}{\sqrt{LC}}$ 2
	$\omega_d = \sqrt{\omega_0^2 - \alpha^2}$
$s_1 = -\alpha + \sqrt{\alpha^2 - \omega_0^2}$	$s_2 = -\alpha - \sqrt{\alpha^2 - \omega_0^2}$

$i_L(t) = C\frac{dV_c(t)}{dt}$

Solve for B_1 B_2

$B_1 = V_c(0) - V_c(\infty)$

$B_2 = \frac{1}{C} i_C(0) + a[V_c(0) - V_c(\infty)]$

$t < 0s$

$V_c(\infty) = 0[V]$

$V_c(0) = 12[V] \cdot \frac{1.2[\Omega]}{1.6[\Omega]} = 9[V]$

$i_C(0) = 0[A]$

$B_1 = 9[V] - 0[V]$

$B_2 = \frac{1}{C}\cdot 0[A] + 6[9 - 0] = 54[V]$

$V_c(t) = (B_1 + B_2 t)e^{-at} + V_c(\infty) = 0.$

$9[V] + 54[V]^{-6t} + 0[V]$

$a = \frac{R_2}{2L} = \frac{1.2[\Omega]}{2(.1[H])} = 6$

$W_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{(.1)(.28[C])}} = 6$

Product Rule

$r' = 54[V] + 54[V]t$

$g' = -6e^{-6t}$ $g = e^{-6t}$

$i_L(t) = .28[F]\ \frac{dV_c}{dt}(9 + 54t)e^{-6t}$

$i_C = C\frac{di(t)}{dt}$

$.28[C]\left[\cancel{-54[V]}e^{-6t} + (\cancel{54[V]}e^{-6t} - 6t \cdot 54[V]t^{-6t})\right]$

$.28[C](-6[V] \cdot 54[V]t\, e^{-6t})$

$= -90.72te^{-6t}$

6.17 A series RLC circuit exhibits the following voltage and current responses:

$$v_C(t) = (6\cos 4t - 3\sin 4t)e^{-2t}\, u(t) \text{ V},$$

$$i_C(t) = -(0.24\cos 4t + 0.18\sin 4t)e^{-2t}\, u(t) \text{ A}.$$

Determine α, ω_0, R, L, and C.

Note: Don't use Ohms Law on a capacitor

underdamp $\to$ $a < w_0$

$D_1 = 6$
$D_2 = 3$ $\to$ $\frac{R}{2L}$ $\quad \frac{1}{\sqrt{LC}}$

$\boxed{a = 2}$

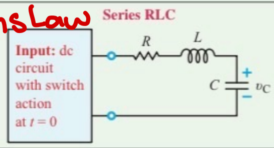
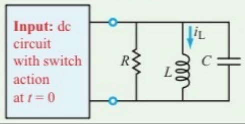
$\boxed{W_0 = 2\sqrt{5}}$

$\boxed{R = 10[\Omega]}$

$\boxed{L = 2.5[H]}$

$\boxed{C = .02[F]}$

Table 6-1: Step response of RLC circuits for $t \geq 0$.

Series RLC	Parallel RLC
	
Total Response	**Total Response**
Overdamped ($\alpha > \omega_0$) $v_C(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t} + v_C(\infty)$ $A_1 = \dfrac{\frac{1}{C} i_C(0) - s_2[v_C(0) - v_C(\infty)]}{s_1 - s_2}$ $A_2 = \dfrac{\frac{1}{C} i_C(0) - s_1[v_C(0) - v_C(\infty)]}{s_2 - s_1}$	**Overdamped** ($\alpha > \omega_0$) $i_L(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t} + i_L(\infty)$ $A_1 = \dfrac{\frac{1}{L} v_L(0) - s_2[i_L(0) - i_L(\infty)]}{s_1 - s_2}$ $A_2 = \dfrac{\frac{1}{L} v_L(0) - s_1[i_L(0) - i_L(\infty)]}{s_2 - s_1}$
Critically Damped ($\alpha = \omega_0$) $v_C(t) = (B_1 + B_2 t)e^{-\alpha t} + v_C(\infty)$ $B_1 = v_C(0) - v_C(\infty)$ $B_2 = \frac{1}{C} i_C(0) + \alpha[v_C(0) - v_C(\infty)]$	**Critically Damped** ($\alpha = \omega_0$) $i_L(t) = (B_1 + B_2 t)e^{-\alpha t} + i_L(\infty)$ $B_1 = i_L(0) - i_L(\infty)$ $B_2 = \frac{1}{L} v_L(0) + \alpha[i_L(0) - i_L(\infty)]$
Underdamped ($\alpha < \omega_0$) $v_C(t) = e^{-\alpha t}(D_1 \cos\omega_d t + D_2 \sin\omega_d t) + v_C(\infty)$ $D_1 = v_C(0) - v_C(\infty)$ $D_2 = \dfrac{\frac{1}{C} i_C(0) + \alpha[v_C(0) - v_C(\infty)]}{\omega_d}$	**Underdamped** ($\alpha < \omega_0$) $i_L(t) = e^{-\alpha t}(D_1 \cos\omega_d t + D_2 \sin\omega_d t) + i_L(\infty)$ $D_1 = i_L(0) - i_L(\infty)$ $D_2 = \dfrac{\frac{1}{L} v_L(0) + \alpha[i_L(0) - i_L(\infty)]}{\omega_d}$
Auxiliary Relations	
$\alpha = \begin{cases} \frac{R}{2L} & \text{Series RLC} \\ \frac{1}{2RC} & \text{Parallel RLC} \end{cases}$	$\omega_0 = \dfrac{1}{\sqrt{LC}}$ $\omega_d = \sqrt{\omega_0^2 - \alpha^2}$
$s_1 = -\alpha + \sqrt{\alpha^2 - \omega_0^2}$	$s_2 = -\alpha - \sqrt{\alpha^2 - \omega_0^2}$

$(W_d)^2 = \left(\sqrt{W_0^2 - a^2}\right)^2 \to$

$W_d^2 = W_0^2 - a^2$
$+a^2 \qquad +a^2$

$\to W_0 = \sqrt{W_d^2 + a^2}$

$\sqrt{4^2 + 2^2} = 2\sqrt{5}$

$D_1 = 6[V] - 0[V]$

$D_2 = \dfrac{\frac{1}{C} i_C(0) + 2[6[V] - 0[V]]}{4}$

$i_C(0) = -.24$

$\dfrac{\frac{1}{C}(-.24) + 12[V]}{4}$

3 unknowns, 3 eq.

$a = \dfrac{R}{2L}$

$W_0 = \dfrac{1}{\sqrt{LC}}$ $\qquad W_d = \sqrt{W_0^2 - a^2}$

$C = .02[C]$

$\dfrac{1}{\sqrt{LC}} = W_0 \to \dfrac{1}{2(\sqrt{L(.02)})} = (2\sqrt{5})^2$

$\dfrac{1}{L(.02)\cdot .02} = 10:02 \to .2$

$\dfrac{1}{L} = .2$

$L = 2.5$

$R = 4L$

$4(2.5) = 10[\Omega]$

6.22 Determine $i_C(t)$ in the circuit of **Fig. P6.22** and plot its waveform for $t \geq 0$.

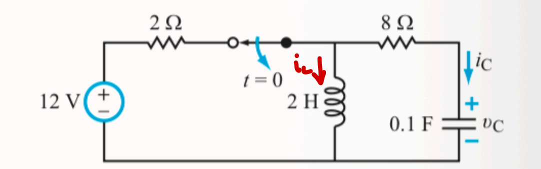
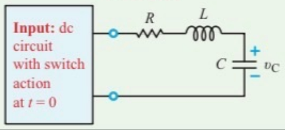
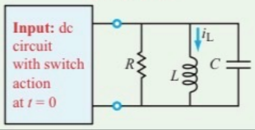


Figure P6.22 Circuit for Problems 6.22 and 6.23.

$\alpha < W_0$

Table 6-1: Step response of RLC circuits for $t \geq 0$.

Series RLC	Parallel RLC
	
Total Response	**Total Response**
Overdamped ($\alpha > \omega_0$)	Overdamped ($\alpha > \omega_0$)
$v_C(t) = A_1e^{s_1t} + A_2e^{s_2t} + v_C(\infty)$	$i_L(t) = A_1e^{s_1t} + A_2e^{s_2t} + i_L(\infty)$
$A_1 = \dfrac{\frac{1}{C}\,i_C(0) - s_2[v_C(0) - v_C(\infty)]}{s_1 - s_2}$	$A_1 = \dfrac{\frac{1}{L}\,v_L(0) - s_2[i_L(0) - i_L(\infty)]}{s_1 - s_2}$
$A_2 = \dfrac{\frac{1}{C}\,i_C(0) - s_1[v_C(0) - v_C(\infty)]}{s_2 - s_1}$	$A_2 = \dfrac{\frac{1}{L}\,v_L(0) - s_1[i_L(0) - i_L(\infty)]}{s_2 - s_1}$
Critically Damped ($\alpha = \omega_0$)	Critically Damped ($\alpha = \omega_0$)
$v_C(t) = (B_1 + B_2t)e^{-\alpha t} + v_C(\infty)$	$i_L(t) = (B_1 + B_2t)e^{-\alpha t} + i_L(\infty)$
$B_1 = v_C(0) - v_C(\infty)$	$B_1 = i_L(0) - i_L(\infty)$
$B_2 = \frac{1}{C}\,i_C(0) + \alpha[v_C(0) - v_C(\infty)]$	$B_2 = \frac{1}{L}\,v_L(0) + \alpha[i_L(0) - i_L(\infty)]$
Underdamped ($\alpha < \omega_0$)	Underdamped ($\alpha < \omega_0$)
$v_C(t) = e^{-\alpha t}(D_1\cos\omega_d t + D_2\sin\omega_d t) + v_C(\infty)$	$i_L(t) = e^{-\alpha t}(D_1\cos\omega_d t + D_2\sin\omega_d t) + i_L(\infty)$
$D_1 = v_C(0) - v_C(\infty)$	$D_1 = i_L(0) - i_L(\infty)$
$D_2 = \dfrac{\frac{1}{C}\,i_C(0) + \alpha[v_C(0) - v_C(\infty)]}{\omega_d}$	$D_2 = \dfrac{\frac{1}{L}\,v_L(0) + \alpha[i_L(0) - i_L(\infty)]}{\omega_d}$

Auxiliary Relations

$$\alpha = \begin{cases} \dfrac{R}{2L} & \text{Series RLC} \\ \dfrac{1}{2RC} & \text{Parallel RLC} \end{cases} \qquad \omega_0 = \frac{1}{\sqrt{LC}} \qquad \omega_d = \sqrt{\omega_0^2 - \alpha^2}$$

$$s_1 = -\alpha + \sqrt{\alpha^2 - \omega_0^2} \qquad s_2 = -\alpha - \sqrt{\alpha^2 - \omega_0^2}$$

Find $V_C(t)$ $t > 0$

$$\alpha = \frac{R}{2L} \rightarrow \frac{8}{2[2]} \rightarrow 2[s^{-1}]$$

$$W_0 = \frac{1}{\sqrt{LC}} \quad \frac{1}{\sqrt{2(.1)}} = 2.236[rad/s]$$

$$W_d = \sqrt{W_0^2 - \alpha^2} \rightarrow \sqrt{2.236^2 - 2^2} = 1$$

$$i_C(t) = C\,\frac{dV_C(t)}{dt}$$

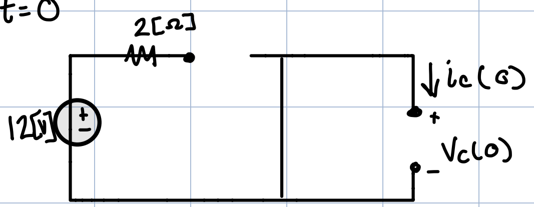
$$V_C(t) = e^{-at}(D_1\cos(W_d)t + D_2\sin(W_d)t) + \underbrace{V_C(\infty)}_{V_C(\infty) = 0[V]}$$

$V_C(\infty) = 0[V] = V_C(0)$

$D_1 = 0[V]$

$$D_2 = \frac{1}{C}\cdot \underbrace{i_C(0)}_{i_C(0) = 6[A]} + \alpha[V_C(0) - V_C(\infty)] \rightarrow \underbrace{\frac{6[A]}{0.1[H]}}_{60[V]} + \cancel{\alpha[V_C(0) - V_C(\infty)]}$$

$t = 0$



$$\frac{12[V]}{2[\Omega]} = 6[A]$$

$$V_C(t) = e^{-2t}(0[V]\cos(1)t + 60[V]\sin(1)t) + 0[V] \rightarrow V_C'(t) = (60[V]\cos t)e^{-2t} - (120[V]\sin t)e^{-2t}$$

$i_C(t) = C \cdot V_C'(t)$

$0.1[F]\,[V_C'(t) = (60[V]\cos t)e^{-2t} - (120[V]\sin t)e^{-2t}]$

$$i_C(t) = (-6[V]\cos(t) + 12[V]\sin(t))e^{-2t}$$

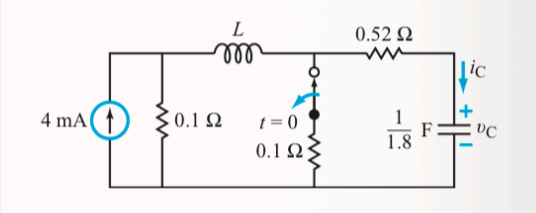
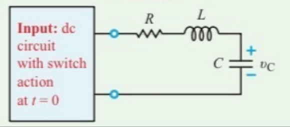
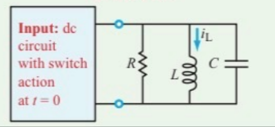


Figure P6.24 Circuit for Problem 6.24 and 6.25.

?

Table 6-1: Step response of RLC circuits for $t \geq 0$.

Series RLC	Parallel RLC
Input: dc circuit with switch action at $t = 0$ 	Input: dc circuit with switch action at $t = 0$ 
Total Response	**Total Response**
Overdamped ($\alpha > \omega_0$)	**Overdamped** ($\alpha > \omega_0$)
$v_C(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t} + v_C(\infty)$	$i_L(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t} + i_L(\infty)$
$A_1 = \dfrac{\frac{1}{C} i_C(0) - s_2[v_C(0) - v_C(\infty)]}{s_1 - s_2}$	$A_1 = \dfrac{\frac{1}{L} v_L(0) - s_2[i_L(0) - i_L(\infty)]}{s_1 - s_2}$
$A_2 = \dfrac{\frac{1}{C} i_C(0) - s_1[v_C(0) - v_C(\infty)]}{s_2 - s_1}$	$A_2 = \dfrac{\frac{1}{L} v_L(0) - s_1[i_L(0) - i_L(\infty)]}{s_2 - s_1}$
Critically Damped ($\alpha = \omega_0$)	**Critically Damped** ($\alpha = \omega_0$)
$v_C(t) = (B_1 + B_2 t)e^{-\alpha t} + v_C(\infty)$	$i_L(t) = (B_1 + B_2 t)e^{-\alpha t} + i_L(\infty)$
$B_1 = v_C(0) - v_C(\infty)$	$B_1 = i_L(0) - i_L(\infty)$
$B_2 = \frac{1}{C} i_C(0) + \alpha[v_C(0) - v_C(\infty)]$	$B_2 = \frac{1}{L} v_L(0) + \alpha[i_L(0) - i_L(\infty)]$
Underdamped ($\alpha < \omega_0$)	**Underdamped** ($\alpha < \omega_0$)
$v_C(t) = e^{-\alpha t}(D_1 \cos \omega_d t + D_2 \sin \omega_d t) + v_C(\infty)$	$i_L(t) = e^{-\alpha t}(D_1 \cos \omega_d t + D_2 \sin \omega_d t) + i_L(\infty)$
$D_1 = v_C(0) - v_C(\infty)$	$D_1 = i_L(0) - i_L(\infty)$
$D_2 = \dfrac{\frac{1}{C} i_C(0) + \alpha[v_C(0) - v_C(\infty)]}{\omega_d}$	$D_2 = \dfrac{\frac{1}{L} v_L(0) + \alpha[i_L(0) - i_L(\infty)]}{\omega_d}$

Auxiliary Relations

$$\alpha = \begin{cases} \dfrac{R}{2L} & \text{Series RLC} \\ \dfrac{1}{2RC} & \text{Parallel RLC} \end{cases} \qquad \omega_0 = \frac{1}{\sqrt{LC}} \qquad \omega_d = \sqrt{\omega_0^2 - \alpha^2}$$

$$s_1 = -\alpha + \sqrt{\alpha^2 - \omega_0^2} \qquad s_2 = -\alpha - \sqrt{\alpha^2 - \omega_0^2}$$

*6.25 Choose the value of the inductor in the circuit of Fig. P6.24 so that v_C exhibits a critically damped response and determine $v_C(t)$ for $t \geq 0$.

$$\alpha = \frac{R}{2L} \qquad \omega_0 = \frac{1}{\sqrt{LC}}$$

$$\frac{R}{2L} = \frac{1}{\sqrt{LC}} \rightarrow \left(\frac{2L}{R}\right)^2 = \left(\frac{\sqrt{LC}}{1}\right)^2$$

$$\frac{4L^2}{\cancel{L}R^2} = \frac{LC}{\cancel{L}} \rightarrow \frac{R^2}{4}\cdot\frac{4L}{R^2} = C \cdot \frac{R^2}{4}$$

$R_{eq} = 0.1[\Omega] + 0.52[\Omega] = 0.62[\Omega]$

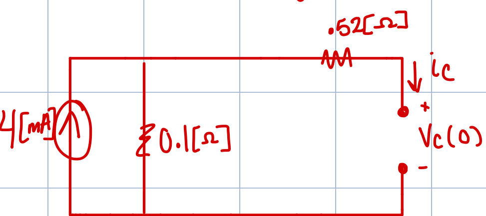
$$L = \frac{C R_{eq}^2}{4} \rightarrow \frac{(0.62[\Omega])^2 (0.556[F])}{4} = 0.0534[H]$$

$$\alpha = \frac{0.62[\Omega]}{2(.0534[H])} = 5.805[s^{-1}] \quad \omega_0 = \frac{1}{\sqrt{LC}} = \cancel{4.24[rad/s]}$$

Since α is = to ω_0 the value of ω_0 will be the same as α

$$B_1 = (V_c(0) - V_c(\infty))[V] = (0.248 - 0.4)[mV] = -0.15[mV]$$

t<0 t>0



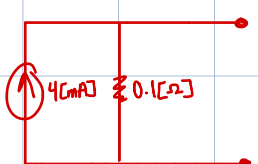
$$B_2 = \frac{1}{C} i_c(0) + \alpha[(0.248 - 0.4)[mV]]$$

$$1.8(2.48[mA]) + 5.805[s^{-1}](-0.15[mV]) = 3.593[mV]$$

$4[mA] \times 0.62[\Omega] = .000248[V]$

$R_{eq} = 0.1[\Omega] + 0.52[\Omega] = 0.62[\Omega] \rightarrow i_c(0) = \frac{V_c(t)}{0.1[\Omega]} \rightarrow \frac{.248[mV]}{0.1[\Omega]} = 2.48[mA]$

t>0



$V_c(\infty) \rightarrow 4[mA] \times 0.1[\Omega] = 0.0004[V]$

$$V_c(t) = (B_1 + B_2 t)e^{-at} + V_c(\infty)$$

$$V_c(t) = \left((-0.248[mV] + 3.593[mV]t)e^{-5.805t} + 0.4[mV]\right)$$

6.29 Choose the value of C in the circuit of Fig. P6.29 so that $v_C(t)$ has a critically damped response for $t \geq 0$. Plot the waveform of $v_C(t)$.

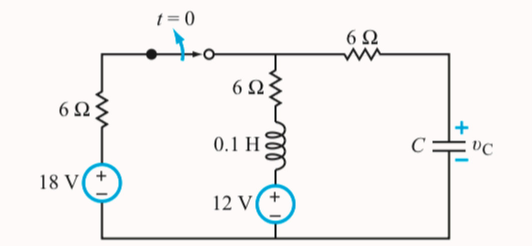
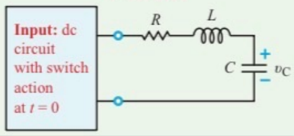
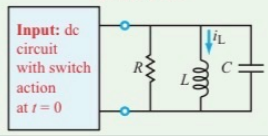


Figure P6.29 Circuit for Problem 6.29.

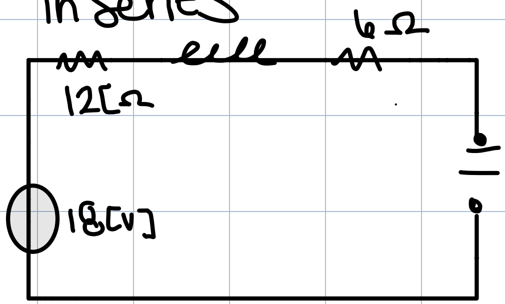
Table 6-1: Step response of RLC circuits for $t \geq 0$.

Series RLC	Parallel RLC
	
Input: dc circuit with switch action at $t = 0$	Input: dc circuit with switch action at $t = 0$
Total Response	**Total Response**
Overdamped ($\alpha > \omega_0$)	Overdamped ($\alpha > \omega_0$)
$v_C(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t} + v_C(\infty)$	$i_L(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t} + i_L(\infty)$
$A_1 = \frac{\frac{1}{C} i_C(0) - s_2[v_C(0) - v_C(\infty)]}{s_1 - s_2}$	$A_1 = \frac{\frac{1}{L} v_L(0) - s_2[i_L(0) - i_L(\infty)]}{s_1 - s_2}$
$A_2 = \frac{\frac{1}{C} i_C(0) - s_1[v_C(0) - v_C(\infty)]}{s_2 - s_1}$	$A_2 = \frac{\frac{1}{L} v_L(0) - s_1[i_L(0) - i_L(\infty)]}{s_2 - s_1}$
Critically Damped ($\alpha = \omega_0$)	Critically Damped ($\alpha = \omega_0$)
$v_C(t) = (B_1 + B_2 t)e^{-\alpha t} + v_C(\infty)$	$i_L(t) = (B_1 + B_2 t)e^{-\alpha t} + i_L(\infty)$
$B_1 = v_C(0) - v_C(\infty)$	$B_1 = i_L(0) - i_L(\infty)$
$B_2 = \frac{1}{C} i_C(0) + \alpha[v_C(0) - v_C(\infty)]$	$B_2 = \frac{1}{L} v_L(0) + \alpha[i_L(0) - i_L(\infty)]$
Underdamped ($\alpha < \omega_0$)	Underdamped ($\alpha < \omega_0$)
$v_C(t) = e^{-\alpha t}(D_1 \cos\omega_d t + D_2 \sin\omega_d t) + v_C(\infty)$	$i_L(t) = e^{-\alpha t}(D_1 \cos\omega_d t + D_2 \sin\omega_d t) + i_L(\infty)$
$D_1 = v_C(0) - v_C(\infty)$	$D_1 = i_L(0) - i_L(\infty)$
$D_2 = \frac{\frac{1}{C} i_C(0) + \alpha[v_C(0) - v_C(\infty)]}{\omega_d}$	$D_2 = \frac{\frac{1}{L} v_L(0) + \alpha[i_L(0) - i_L(\infty)]}{\omega_d}$
Auxiliary Relations	
$\alpha = \begin{cases} \frac{R}{2L} & \text{Series RLC} \\ \frac{1}{2RC} & \text{Parallel RLC} \end{cases}$	$\omega_0 = \frac{1}{\sqrt{LC}}$, $\omega_d = \sqrt{\omega_0^2 - \alpha^2}$
$s_1 = -\alpha + \sqrt{\alpha^2 - \omega_0^2}$	$s_2 = -\alpha - \sqrt{\alpha^2 - \omega_0^2}$

Part a) Find C

C from critically damp

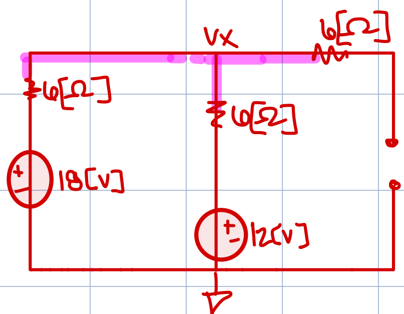
In Series

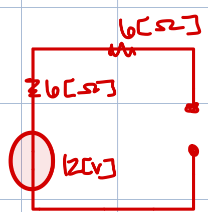


$$\frac{R}{2L} = \frac{1}{\sqrt{LC}} \rightarrow \text{Solving for } C = \frac{4L}{R^2} = \frac{4(.1)}{(12)^2} = 2.778[mF]$$

Req= 6[Ω] || 6[Ω] + 6[Ω] = 9[Ω]

Part b) $t<0$ $t>0$





$V_c(\infty) = 12[V]$

$i_c(0) = \frac{12[V]}{6[Ω]} = -.5[..]$

$i_c(0) + .5[A] = 0$

$V_c(0^-) = 15[V] \leftarrow V_c(0) = 15[V]$

$$\frac{V_x - 18[V]}{6[Ω]} - \frac{V_x - 12[V]}{6[Ω]} + \cancel{\frac{V_x}{6[Ω]}} = 0$$

$\alpha = \frac{R}{2L} \rightarrow \frac{12[Ω]}{2(0.1[H])} = 60[s^{-1}] = \omega_0 = 60[rad/s]$

$V_c(t) = (B_1 + B_2 t)e^{-\alpha t} + V_c(\infty)$

$B_1 = 15[V] - 12[V] = 3[V]$

$B_2 = \frac{1}{C} i_c(0) + \alpha[V_c(0) - V_c(\infty)]$

$B_2 \frac{-0.5[A]}{2.778[mF]} + 60[3[V]] = 0$

$B_2 = 14.399[mV]$

$V_c(t) = ((3[V] + 14.39[mV]t)e^{-60t} + 12[V])$

*6.32 For the circuit in Fig. P6.32, assume that before $t = 0$, the circuit had been in that state for a long time. Find $v_C(t)$ and $i_L(t)$ for $t \geq 0$.

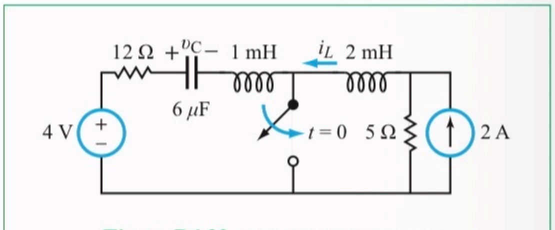
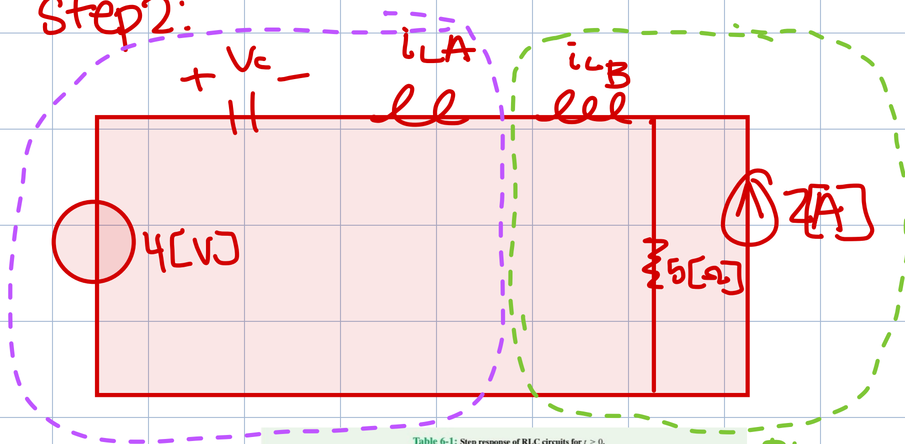


Figure P6.32 Circuit for Problem 6.32.

Step2:



Step1:

$+ < 0[V]$

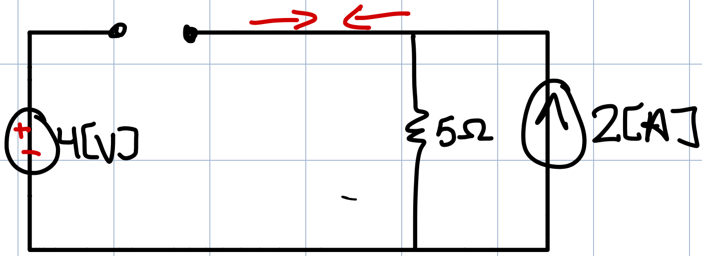
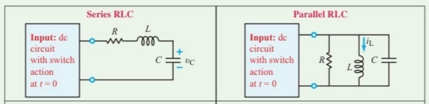


Table 6-1: Step response of RLC circuits for $t \geq 0$.



Series RLC	Parallel RLC
Total Response	**Total Response**
Overdamped ($\alpha > \omega_0$)	**Overdamped** ($\alpha > \omega_0$)
$v_C(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t} + v_C(\infty)$	$i_L(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t} + i_L(\infty)$
$A_1 = \frac{\frac{1}{C} i_C(0) - s_2[v_C(0) - v_C(\infty)]}{s_1 - s_2}$	$A_1 = \frac{\frac{1}{L} v_L(0) - s_2[i_L(0) - i_L(\infty)]}{s_1 - s_2}$
$A_2 = \frac{\frac{1}{C} i_C(0) - s_1[v_C(0) - v_C(\infty)]}{s_2 - s_1}$	$A_2 = \frac{\frac{1}{L} v_L(0) - s_1[i_L(0) - i_L(\infty)]}{s_2 - s_1}$
Critically Damped ($\alpha = \omega_0$)	**Critically Damped** ($\alpha = \omega_0$)
$v_C(t) = (B_1 + B_2 t)e^{-\alpha t} + v_C(\infty)$	$i_L(t) = (B_1 + B_2 t)e^{-\alpha t} + i_L(\infty)$
$B_1 = v_C(0) - v_C(\infty)$	$B_1 = i_L(0) - i_L(\infty)$
$B_2 = \frac{1}{C} i_C(0) + \alpha[v_C(0) - v_C(\infty)]$	$B_2 = \frac{1}{L} v_L(0) + \alpha[i_L(0) - i_L(\infty)]$
Underdamped ($\alpha < \omega_0$)	**Underdamped** ($\alpha < \omega_0$)
$v_C(t) = e^{-\alpha t}(D_1 \cos \omega_d t + D_2 \sin \omega_d t) + v_C(\infty)$	$i_L(t) = e^{-\alpha t}(D_1 \cos \omega_d t + D_2 \sin \omega_d t) + i_L(\infty)$
$D_1 = v_C(0) - v_C(\infty)$	$D_1 = i_L(0) - i_L(\infty)$
$D_2 = \frac{\frac{1}{C} i_C(0) + \alpha[v_C(0) - v_C(\infty)]}{\omega_d}$	$D_2 = \frac{\frac{1}{L} v_L(0) + \alpha[i_L(0) - i_L(\infty)]}{\omega_d}$
Auxiliary Relations	
$\alpha = \begin{cases} \frac{R}{2L} & \text{Series RLC} \\ \frac{1}{2RC} & \text{Parallel RLC} \end{cases}$	$\omega_0 = \frac{1}{\sqrt{LC}}$
	$\omega_d = \sqrt{\omega_0^2 - \alpha^2}$
$s_1 = -\alpha + \sqrt{\alpha^2 - \omega_0^2}$	$s_2 = -\alpha - \sqrt{\alpha^2 - \omega_0^2}$

$V_c(0^-) = -4[V] + V_c(0^-) + 10[V] = 0$

$V_c(0^-) = -6[V]$

$i_{LA}(0^-) = 0[A] = i_{LB}(0^-)$

$\alpha = \frac{R}{2L} < w_0 = \frac{1}{\sqrt{LC}}$

$\frac{12[\Omega]}{2(1[mH])} = 6000[s^{-1}]$

$\frac{1}{\sqrt{(1[mH])(6[\mu F])}} = 12909.9[rad/s]$

Step3: RLC

Underdamp formula

$V_c(t) = e^{-6000[rad/s]t}\left((-10[V])\cos(11430.95t) - 5.249[V]\sin(11430.95t)\right) + 4[V]$

Step6:

$D_1 = (-6[V] - 4[V]) = -10[V]$

Step7:

$D_2 = \frac{6000[-10]}{11430.95} = -5.249[V]$

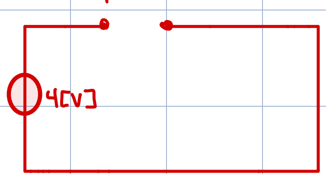
Step5:

$V_c(\infty) = 4[V]$

Step4:

$\omega_d = \sqrt{\omega_0^2 - \alpha^2} \Rightarrow \sqrt{6000^2 - 12909.9} = 11430.95[rad/s]$

$+ V_c(\infty)$



$V_c(\infty) = 4[V]$

Because the cap. is in series with the inductor.

$i_C(0)$

$i_C(0) = 0[A]$

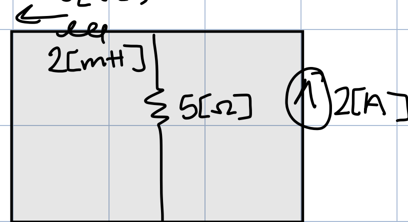
RL

$i_L(t)$

$i_L(0^-) = 0[A]$

$i_L(\infty) = 2[A]$

$i_L(t)$



$\tau = \frac{2[mH]}{5[\Omega]} = .4[mA]$

$2[A] - 2[A]e^{-\frac{t}{.4[mA]}}$ for $t \geq 0$

*6.35 For the circuit in **Fig. P6.35**, find $v_C(t)$ for $t \geq 0$.

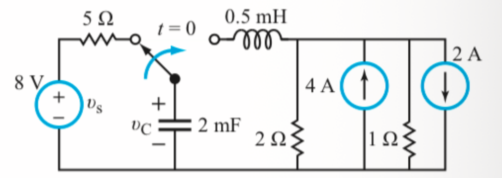
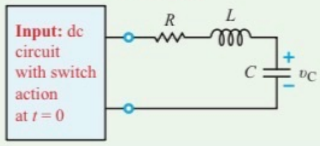
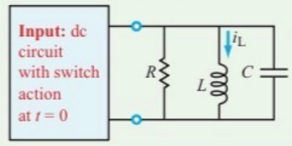


Figure P6.35 Circuit for Problem 6.35.

Table 6-1: Step response of RLC circuits for $t \geq 0$.

Series RLC	Parallel RLC
	
Total Response	**Total Response**
Overdamped ($\alpha > \omega_0$)	**Overdamped** ($\alpha > \omega_0$)
$v_C(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t} + v_C(\infty)$	$i_L(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t} + i_L(\infty)$
$A_1 = \dfrac{\frac{1}{C} i_C(0) - s_2[v_C(0) - v_C(\infty)]}{s_1 - s_2}$	$A_1 = \dfrac{\frac{1}{L} v_L(0) - s_2[i_L(0) - i_L(\infty)]}{s_1 - s_2}$
$A_2 = \dfrac{\frac{1}{C} i_C(0) - s_1[v_C(0) - v_C(\infty)]}{s_2 - s_1}$	$A_2 = \dfrac{\frac{1}{L} v_L(0) - s_1[i_L(0) - i_L(\infty)]}{s_2 - s_1}$
Critically Damped ($\alpha = \omega_0$)	**Critically Damped** ($\alpha = \omega_0$)
$v_C(t) = (B_1 + B_2 t)e^{-\alpha t} + v_C(\infty)$	$i_L(t) = (B_1 + B_2 t)e^{-\alpha t} + i_L(\infty)$
$B_1 = v_C(0) - v_C(\infty)$	$B_1 = i_L(0) - i_L(\infty)$
$B_2 = \frac{1}{C} i_C(0) + \alpha[v_C(0) - v_C(\infty)]$	$B_2 = \frac{1}{L} v_L(0) + \alpha[i_L(0) - i_L(\infty)]$
Underdamped ($\alpha < \omega_0$)	**Underdamped** ($\alpha < \omega_0$)
$v_C(t) = e^{-\alpha t}(D_1 \cos \omega_d t + D_2 \sin \omega_d t) + v_C(\infty)$	$i_L(t) = e^{-\alpha t}(D_1 \cos \omega_d t + D_2 \sin \omega_d t) + i_L(\infty)$
$D_1 = v_C(0) - v_C(\infty)$	$D_1 = i_L(0) - i_L(\infty)$
$D_2 = \dfrac{\frac{1}{C} i_C(0) + \alpha[v_C(0) - v_C(\infty)]}{\omega_d}$	$D_2 = \dfrac{\frac{1}{L} v_L(0) + \alpha[i_L(0) - i_L(\infty)]}{\omega_d}$
Auxiliary Relations	
$\alpha = \begin{cases} \dfrac{R}{2L} & \text{Series RLC} \\ \dfrac{1}{2RC} & \text{Parallel RLC} \end{cases}$	$\omega_0 = \dfrac{1}{\sqrt{LC}}$, $\omega_d = \sqrt{\omega_0^2 - \alpha^2}$
$s_1 = -\alpha + \sqrt{\alpha^2 - \omega_0^2}$	$s_2 = -\alpha - \sqrt{\alpha^2 - \omega_0^2}$

Step 1. Solve for α & W_0

this will help you know where the circuit is in the formulas.

in series $\quad R_{eq} = 2[\Omega] \,//\, 1[\Omega]$

$$\alpha = \frac{R}{2L} = \frac{.6667[\Omega]}{2(.5[mH])} = 666.67[C]$$

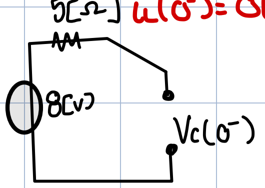
$$W_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{(.5[mH])(2[mF])}} = 1000$$

$\alpha < W_0$ = Underdamp.

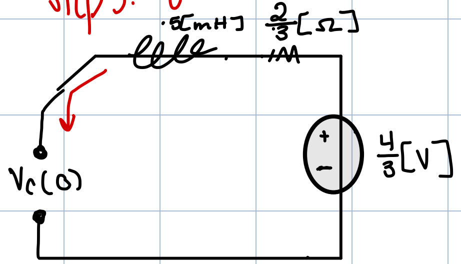
Step 2: $t < 0$

$V_c(0^-) = 8[V]$

$i_L(0^-) = 0[A]$



Step 3: t



$V_c(0) = 8[V]$

Step 4: $t \geq 0$

$V_c(\infty) = 4/3[V]$

$$2[A] \cdot \frac{(2 \cdot 1)}{(2+1)}[\Omega]$$

$$V_c(t) = e^{-666.67t}\left(6.667[V]\cos(745.35t) + 5.963\sin(745.35t)\right) + 1.33[V]$$

$D_1 = 8 - 4/3 = 6.667[V]$

$$D_2 = \frac{\alpha[V_0 - V_\infty]}{W_d} \rightarrow \sqrt{W_0^2 - \alpha^2} = 745.35$$

$$V_c'(0) = \frac{i_c(0)}{C}$$

$i_c(0) = i_L(0) = 0$

$$\frac{666.67(6.667)}{745.35} = 5.963[V]$$

*6.40 Determine $i_L(t)$ in the circuit of Fig. P6.40 and plot its waveform for $t \geq 0$.

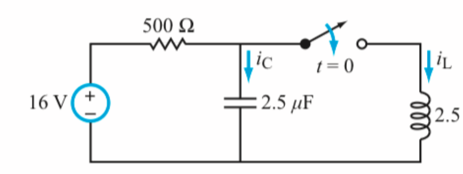
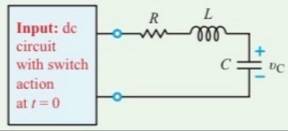
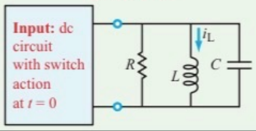


Figure P6.40 Circuit for Problems 6.40 and 6.41.

$\alpha = \omega_0$

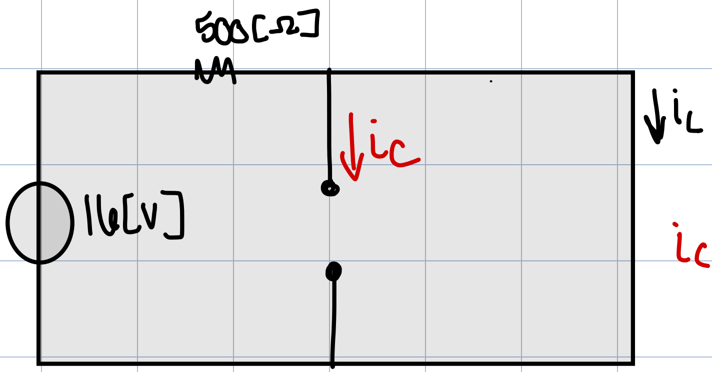
critically damp.

Table 6-1: Step response of RLC circuits for $t \geq 0$.

Series RLC	Parallel RLC
	
Total Response	**Total Response**
Overdamped ($\alpha > \omega_0$)	**Overdamped** ($\alpha > \omega_0$)
$v_C(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t} + v_C(\infty)$	$i_L(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t} + i_L(\infty)$
$A_1 = \frac{\frac{1}{C} i_C(0) - s_2[v_C(0) - v_C(\infty)]}{s_1 - s_2}$	$A_1 = \frac{\frac{1}{L} v_L(0) - s_2[i_L(0) - i_L(\infty)]}{s_1 - s_2}$
$A_2 = \frac{\frac{1}{C} i_C(0) - s_1[v_C(0) - v_C(\infty)]}{s_2 - s_1}$	$A_2 = \frac{\frac{1}{L} v_L(0) - s_1[i_L(0) - i_L(\infty)]}{s_2 - s_1}$
Critically Damped ($\alpha = \omega_0$)	**Critically Damped** ($\alpha = \omega_0$)
$v_C(t) = (B_1 + B_2 t)e^{-\alpha t} + v_C(\infty)$	$i_L(t) = (B_1 + B_2 t)e^{-\alpha t} + i_L(\infty)$
$B_1 = v_C(0) - v_C(\infty)$	$B_1 = i_L(0) - i_L(\infty)$
$B_2 = \frac{1}{C} i_C(0) + \alpha[v_C(0) - v_C(\infty)]$	$B_2 = \frac{1}{L} v_L(0) + \alpha[i_L(0) - i_L(\infty)]$
Underdamped ($\alpha < \omega_0$)	**Underdamped** ($\alpha < \omega_0$)
$v_C(t) = e^{-\alpha t}(D_1 \cos \omega_d t + D_2 \sin \omega_d t) + v_C(\infty)$	$i_L(t) = e^{-\alpha t}(D_1 \cos \omega_d t + D_2 \sin \omega_d t) + i_L(\infty)$
$D_1 = v_C(0) - v_C(\infty)$	$D_1 = i_L(0) - i_L(\infty)$
$D_2 = \frac{\frac{1}{C} i_C(0) + \alpha[v_C(0) - v_C(\infty)]}{\omega_d}$	$D_2 = \frac{\frac{1}{L} v_L(0) + \alpha[i_L(0) - i_L(\infty)]}{\omega_d}$
Auxiliary Relations	
$\alpha = \begin{cases} \frac{R}{2L} & \text{Series RLC} \\ \frac{1}{2RC} & \text{Parallel RLC} \end{cases}$	$\omega_0 = \frac{1}{\sqrt{LC}}$; $\omega_d = \sqrt{\omega_0^2 - \alpha^2}$
$s_1 = -\alpha + \sqrt{\alpha^2 - \omega_0^2}$	$s_2 = -\alpha - \sqrt{\alpha^2 - \omega_0^2}$

$$\alpha = \frac{1}{2RC} \rightarrow \frac{1}{2(500[\Omega])(2.5[\mu F])} = 400[s^{-1}]$$

$$W_0 = \frac{1}{\sqrt{LC}} \rightarrow \frac{1}{\sqrt{(2.5\mu F)(2.5H)}} \rightarrow 400[rad/s]$$



$$i_C(0) = \frac{16[V]}{2.5\mu F} = 6.4[A]$$

?

$$v_L(0) = 16[V]$$

$$i_L(0) = 0[A]$$

$$i_L(\infty) = \frac{16[V]}{500[\Omega]} = 32[mA]$$

$$B_1 = 0[A] - 32[mA] = -32[mA]$$

$$B_2 = \frac{16[V]}{2.5} + 400[s^{-1}](-32[mA]) = -6.4[A]$$

$$i_L(t) = [(-32[mA] - 6.4(t)[A])e^{-400t} + 32[mA])$$

*6.46 In the circuit in Fig. P6.46, $\upsilon_s = 20$ V.

(a) Determine $i_L(t)$ for $t \geq 0$.

(b) If the source is changed to $\upsilon_s(t) = e^{-2t}\, u(t)$, can you still use the solution method in part (a) to find $i_L(t)$? If not, why not?

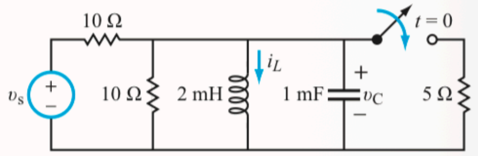
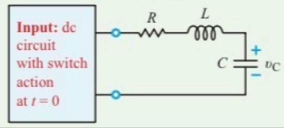
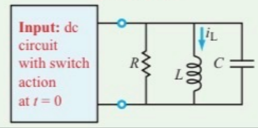


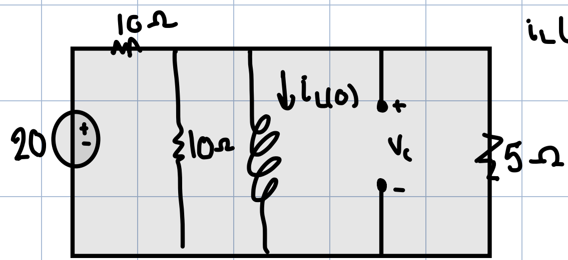
Figure P6.46 Circuit for Problem 6.46.

Series RLC	Parallel RLC
 Input: dc circuit with switch action at $t=0$	 Input: dc circuit with switch action at $t=0$
Total Response	**Total Response**
Overdamped ($\alpha > \omega_0$)	**Overdamped** ($\alpha > \omega_0$)
$\upsilon_C(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t} + \upsilon_C(\infty)$	$i_L(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t} + i_L(\infty)$
$A_1 = \dfrac{\frac{1}{C} i_C(0) - s_2[\upsilon_C(0) - \upsilon_C(\infty)]}{s_1 - s_2}$	$A_1 = \dfrac{\frac{1}{L} \upsilon_L(0) - s_2[i_L(0) - i_L(\infty)]}{s_1 - s_2}$
$A_2 = \dfrac{\frac{1}{C} i_C(0) - s_1[\upsilon_C(0) - \upsilon_C(\infty)]}{s_2 - s_1}$	$A_2 = \dfrac{\frac{1}{L} \upsilon_L(0) - s_1[i_L(0) - i_L(\infty)]}{s_2 - s_1}$
Critically Damped ($\alpha = \omega_0$)	**Critically Damped** ($\alpha = \omega_0$)
$\upsilon_C(t) = (B_1 + B_2 t)e^{-\alpha t} + \upsilon_C(\infty)$	$i_L(t) = (B_1 + B_2 t)e^{-\alpha t} + i_L(\infty)$
$B_1 = \upsilon_C(0) - \upsilon_C(\infty)$	$B_1 = i_L(0) - i_L(\infty)$
$B_2 = \frac{1}{C} i_C(0) + \alpha[\upsilon_C(0) - \upsilon_C(\infty)]$	$B_2 = \frac{1}{L} \upsilon_L(0) + \alpha[i_L(0) - i_L(\infty)]$
Underdamped ($\alpha < \omega_0$)	**Underdamped** ($\alpha < \omega_0$)
$\upsilon_C(t) = e^{-\alpha t}(D_1 \cos\omega_d t + D_2 \sin\omega_d t) + \upsilon_C(\infty)$	$i_L(t) = e^{-\alpha t}(D_1 \cos\omega_d t + D_2 \sin\omega_d t) + i_L(\infty)$
$D_1 = \upsilon_C(0) - \upsilon_C(\infty)$	$D_1 = i_L(0) - i_L(\infty)$
$D_2 = \dfrac{\frac{1}{C} i_C(0) + \alpha[\upsilon_C(0) - \upsilon_C(\infty)]}{\omega_d}$	$D_2 = \dfrac{\frac{1}{L} \upsilon_L(0) + \alpha[i_L(0) - i_L(\infty)]}{\omega_d}$
Auxiliary Relations	
$\alpha = \begin{cases} \frac{R}{2L} & \text{Series RLC} \\ \frac{1}{2RC} & \text{Parallel RLC} \end{cases}$	$\omega_0 = \dfrac{1}{\sqrt{LC}}$ $\omega_d = \sqrt{\omega_0^2 - \alpha^2}$
$s_1 = -\alpha + \sqrt{\alpha^2 - \omega_0^2}$	$s_2 = -\alpha - \sqrt{\alpha^2 - \omega_0^2}$

$$\alpha = \frac{1}{2RC} \rightarrow \frac{1}{2(20\Omega)(1e\text{-}3)} = 25\,[s^{-1}]$$

$$\omega_0 = \frac{1}{\sqrt{LC}} \rightarrow \frac{1}{\sqrt{2mH \cdot 1mF}} = 707.107\,[rad/s]$$

$\alpha < \omega_0 \leftarrow$ under damp.



$$i_L(0) = \frac{V_s}{R_{eq}} \rightarrow \frac{20[V]}{10[\Omega]} \rightarrow 2[A] \qquad V_L(0) = 20[V]$$

a) $i_L(t) = 2[A]$

b) N/A, only works for dc sources.